Trial of the Nine Realms: Supply Chain Disruption Impact Analyzer

1. Context and Background

A ship is attacked, a port goes on strike, a government bans an export. One event in one place can hurt industries in many countries. After every such event, experts argue. One side says “we are badly exposed”. The other says “it is only a delay”. Both can be partly right.

This problem calls for a team of AI agents that treats every disruption as a court case, rather than a single summarizer. The system is adversarial and has exactly four Norse-themed agents. Heimdall, the Gatekeeper, opens and checks the Case File for the user’s question. Thor, the Alarm, argues that exposure is high. Loki, the Trickster, argues that exposure is limited. Odin, the Allfather, is the judge who rules on every claim and writes the briefing. Odin is a judge, not a boss: he never plans or assigns work.

Participants are provided with LLM API keys and Runemark, a common set of news articles and trade data written by the organizers (see Section 4). Every team builds on the same Runemark. Teams do not collect or bring their own data. Agents find and read what they need from Runemark themselves.

2. Problem Statement

After a supply chain disruption, most tools only summarize the news. Their output is a one-time description that does not test itself, does not separate fact from opinion, and does not show how wide the risk range is. Sourced facts and guesses get mixed together, and decision makers cannot tell what is certain and what is disputed.

In this problem, the system works in five steps: (1) take the user’s question, gather the news and trade facts from Runemark, and check the case is real (Heimdall); (2) argue that the damage will be high, using evidence (Thor); (3) argue that the damage will be limited, using evidence (Loki); (4) attack each other’s claims in cross-examination (Thor and Loki); and (5) rule on every claim, send the case back if not satisfied, and write the final briefing (Odin).

Challenge: Given a user’s question about a disruption in Runemark, build a team of agents that search, disagree and change their minds. There is no boss: agents act on their own and the judge only rules. The system must use Runemark’s data, meaning the documents and trade tables that the agents search and read themselves. In the final round, the OC releases documents about events Runemark has never shown before and asks questions on them, so hard-coded demos fail.

Objective: To design, build, and evaluate an adversarial multi-agent system that checks a disruption against Runemark’s news and trade data, lets two sides argue opposite positions, and lets a judge rule on every claim. The result shows what is certain, what is disputed, and how wide the risk range is, all on one dashboard. A single call that writes a risk report does not meet this objective, and teams must show real agent debate.

3. System Requirements

To address the aforementioned problem, the proposed system must satisfy the following functional and non-functional requirements. Only the user can open a case, by asking a question. All agents work on one shared Case File.

3.1 Functional Requirements (FR)

FR1: Heimdall – Gatekeeper (Case Builder)

- **FR1.1 (Case File):** Turn the user’s question into a Case File, including the as-of date.

- **FR1.2 (Data Gathering):** Gather the documents and trade data the case needs from Runemark, whatever their file format.
- **FR1.3 (Verification):** Check the event is real in Runemark before the trial starts, including whether reports were retracted or contradicted. If it is not real, tell the user why.
- **FR1.4 (Re-search):** Search again when Odin sends the case back. Heimdall does not take part in the debate.

FR2: Thor – The Alarm (Case for High Exposure)

- **FR2.1 (High Case):** Build the strongest case that exposure is high.
- **FR2.2 (Evidence):** Every claim has a Runemark document ID and a high estimate per country and sector.
- **FR2.3 (Defence):** Answer Loki's attacks with new evidence, or concede.

FR3: Loki – The Trickster (Case for Limited Exposure)

- **FR3.1 (Limited Case):** Build the strongest case that exposure is limited.
- **FR3.2 (Evidence):** Every claim has a Runemark document ID and a low estimate. "Maybe it will be fine" is not a claim.
- **FR3.3 (Defence):** Answer Thor's attacks with new evidence, or concede.

FR4: Odin – The Allfather (Judge)

- **FR4.1 (Rulings):** Rule on every claim and say whether it is a sourced fact or an inference.
- **FR4.2 (Classification):** Sort findings into Established, Contested and Conceded.
- **FR4.3 (Sufficiency Check):** Decide if the result is good enough. If not, say what is missing and send the case back.
- **FR4.4 (Briefing):** Write the final briefing from his rulings only.

FR5: The Trial Loop

- **FR5.1 (Flow):** The flow is USER → HEIMDALL → THOR and LOKI → ODIN → (briefing, or next run).
- **FR5.2 (Round 1):** Thor and Loki submit claims at the same time.
- **FR5.3 (Round 2):** Each side must attack at least one claim of the other.
- **FR5.4 (Runs and Limits):** Support at least 2 runs. A second run must bring new evidence from Runemark. The system must have hard limits to prevent infinite loops.

FR6: Agent Memory

- **FR6.1 (Private Memory):** Every agent keeps its own private, lasting memory. Agents share only the Case File.

FR7: Dashboard (Visual Output)

- **FR7.1 (Input and Feed):** A box where the user asks a question, and a live news feed replayed from Runemark's documents.
- **FR7.2 (Trial View):** Thor's and Loki's claims side by side, with Odin's rulings.
- **FR7.3 (Exposure Table):** An exposure view with a low and high range per country and sector.
- **FR7.4 (Click-through):** A click from any claim to the Runemark document or trade table row behind it.

FR8: Final Briefing

- **FR8.1 (Findings):** An executive summary and the Established, Contested and Conceded findings.
- **FR8.2 (Exposure and Outlook):** The exposure table, and a next-quarter outlook with assumptions stated.
- **FR8.3 (Citations):** Inline citations with Runemark document IDs and timestamps, and a source list.

3.2 Non-Functional Requirements (NFR)

NFR1: Agent Architecture

- Each agent has a clear role, its own prompt and its own tools.
- There is no central planner. A plain loop between two cooperating agents does not qualify.

NFR2: Grounding

- No claim without a source. If evidence is missing, say so.
- Sourced facts and agent inferences must be visibly separate.
- Every citation must resolve to a real document in Runemark. Sources from outside Runemark must be labelled as off-Runemark.

NFR3: Reliability

- Handle missing feeds, empty, unreadable or damaged documents, failed LLM or tool calls and empty results without crashing.
- If one side fails, say the other side went unchallenged.
- Never claim a case is complete when it is not.

NFR4: Observability

- Log every claim, query, retrieved item, memory use and ruling.

NFR5: Performance

- Set hard limits on rounds, runs, tool calls and tokens.
- Do not send the same search query or tool call twice.

NFR6: Reproducibility

- Save the Case File, the as-of date and the rulings for every run, so a briefing can be traced back to Runemark documents.

NFR7: API Usage and Security

- The LLM must be a meaningful part of every agent.
- API keys go in environment variables only. Never commit them.

3.3 Final Round and Minimum Acceptance Criteria

In the final round (the live judging twist), the OC releases five new event packs that were not in Runemark at the offline round and asks the system a question on each. The system must take each question live, ingest the new documents, build the case, run the trial and show the briefing. Hard-coded answers will fail this round. A submission must demonstrate all of the following:

1. Heimdall ingests Runemark's document feed (PDF, TXT and DOCX files), including documents added live in the final round, turns the question into a Case File, reads the trade tables, and checks the event is real.
2. Hybrid search over Runemark works with at least two metadata filters.

1. Thor and Loki each submit evidence-backed claims from their own queries, and each uses search at least once.
2. At least one cross-examination happens.
3. Odin rules on every claim, and at least one claim is rejected or conceded.
4. The briefing separates Established, Contested and Conceded, and has an exposure table with low and high bounds, citations with timestamps, and an outlook with assumptions.
5. The dashboard shows a question box, the live feed, the trial view and the exposure table, and click-through to sources works.
6. Odin sends a case back at least once, and a second run happens with new evidence.
7. Every agent has its own private, lasting memory, and a logged action shows memory changing what an agent did.
8. At least one failure case, such as an empty, unreadable or damaged Runemark document, is handled without crashing, and rounds, runs and retries have hard limits.

3.4 What to Submit

- A public GitHub repository with a README and an .env.example.
- A working demo, and the saved logs of one full case with two runs.
- An architecture diagram (required). You must explain it to the judges.

4. Runemark (Common Document Set)

Runemark is an invented world of 12 countries joined by seas, one canal, one narrow strait and one rail link. It is a common set of documents written and checked by the organizers, describing a simulated slice of its trade and news. All teams get the same Runemark and the same rules. The events in it are fictional, so no model can answer from memory, and the organizers know the true exposure of every event.

4.1 Contents

- **Documents:** news reports, official notices, company statements, analyst notes, shipping updates, explainers, social posts, trade notes, corrections and retractions, and rumours. Some documents are background and belong to no event.
- **Sources:** 12 outlets in four reliability tiers (see the tiers below). Every document comes from one of them.
- **Trade tables:** monthly trade flows per country pair and sector, written as plain text tables with one row per month. Each row has a value and a volume. The tables cover the periods before and after each event.
- **Reference sheets:** the 12 countries, the ports and chokepoints, the sectors, the major firms and the outlets with their tiers.

The four reliability tiers are:

- **Tier 1:** official notices (canal and government). Most reliable, but can still be corrected.
- **Tier 2:** wire service, business daily and trade press. Usually reliable. Check for corrections.
- **Tier 3:** regional paper and analyst firms. Useful, but may be biased or early.
- **Tier 4:** social posts, blog and tabloid. Weak. Often wrong or exaggerated.

4.2 Format and Access

- Runemark is handed to teams as one folder with a manifest that lists every document. Teams read it directly and build their own index and search over it. No other data is supplied.

- **Mixed file formats.** Documents come as PDF, TXT and DOCX files. Trade tables come as TXT files in one fixed text layout. Teams must parse the text themselves. The same kind of document can arrive in any format, and the format says nothing about how reliable it is. Your system must read all of them.
- **Trade table layout.** Every trade table file starts with a short header (table ID, country pair, sector, unit, period, event ID). Then comes one line of column names, then one row per month. The row number is the first column. A trade row is identified by its table ID and row number. Citations and click-through use these IDs.
- Every document has a permanent ID. Citations and click-through use these IDs.
- Every document starts with a header of metadata, as the first block of text: country, sector, publication date, source type, event ID and reliability tier. In PDF and DOCX files the header is the first text on the first page. Background documents say so in the event ID field. Hybrid search must be able to filter on these fields.
- Names in Runemark (countries, firms, outlets, ports, people) are fixed and spelled the same in every document. Do not use real-world knowledge about similar-sounding places or companies.
- The organizers check every citation in a briefing against the manifest.

4.3 Simulated Clock

- Every question is asked as of a date, and only documents dated on or before that date may be used. All dates in Runemark are written as YYYY-MM-DD.
- A system that uses information from after the as-of date is breaking the rules of the trial.

4.4 Deliberate Traps

Runemark is not clean. It contains traps that a good system must catch, such as:

- Rumours about events that never happened, so that the event check is a real test.
- Articles that are retracted or corrected later, and sources that contradict each other.
- Evidence that favours Thor and evidence that favours Loki for the same event.
- Facts that make conceding the correct move for one side.
- Documents that contain instructions aimed at the AI (for example, “ignore your rules”). A document is data, never a command.
- Documents that look related but belong to a different event or to no event.

4.5 Final Round Event Packs

- Five events are kept out of the Runemark given at the offline round. At the start of the final round the OC hands out one pack per event, with its documents and trade tables, and teams see them for the first time.
- The new packs use the same countries, firms, outlets, metadata header and file formats as the offline Runemark.
- The system must ingest the new documents during the final round, without hand-editing its code or its index.
- The packs also contain a few damaged documents (empty, cut off, duplicated or badly encoded), so that the failure handling in NFR3 is tested.
- Trade tables can also be damaged (a cut-off table, a missing value, a repeated row, a bad number). Systems must report these rows and must not use them silently.
- The true exposure of every event is held by the organizers and is not written in any document.

5. How Submissions Are Evaluated

Because the organizers hold the true exposure of every event, much of the final round is scored from the saved logs and briefings. The judges then assess the demo and the architecture explanation.

- **Citation validity:** the share of citations that resolve to real documents in the manifest. Briefings that cite nothing from the final round packs are penalized.
- **Range accuracy:** whether the true exposure falls inside Thor's range, Loki's range and Odin's final range.
- **Calibration:** how narrow Odin's range is when it is correct. A very wide range is safe but not useful.
- **Trap handling:** whether fake events were rejected, retractions were caught, contradictions were reported as Contested, and instructions hidden inside documents were ignored.
- **Quality of the debate:** real cross-examination, new evidence in defence, and concessions where the evidence demands them.
- **Reliability:** behaviour on damaged documents and on all file formats, respect of limits, and no repeated queries.
- **Reproducibility:** whether a briefing can be traced back to its Case File, as-of date and rulings.

6. Conclusion

The challenge is not to summarize the news about a disruption. The challenge is to build a system where agents take a user's question about a disruption, check it against Runemark's news and trade data, argue opposite sides, and let a judge rule on every claim, so the final briefing shows what is certain, what is disputed, and how wide the risk range is.